The automotive display ecosystem influences industries

Developing automotive displays involves various crucial components, and the selection may vary based on the display technology used. Here are the primary components integral to the creation of automotive displays:

- ***Cover glass.*** Essential for protecting the display and optimising visibility while minimising glare and reflections. It must be durable and resilient to withstand various environmental conditions
- ***Diffuser.*** Often used in LCDs to distribute light evenly across the screen, ensuring consistent brightness and colour uniformity
- ***LCD technology.*** Liquid crystal display (LCD) technology is the foundation for many automotive displays, comprising layers of liquid crystals and polarizers to manipulate light for image generation
- ***Backlight unit.*** Necessary for illuminating LCDs, with light emitting diodes (LEDs) being a common choice to provide the required illumination
- ***Colour conversion film.*** A colour conversion film is applied in LCDs to achieve precise colour representation, contributing to vibrant and accurate visuals
- ***Adhesives.*** Due to the demanding automotive environment, high-strength adhesives are crucial for securely bonding components, ensuring the display's integrity under vibrations, temperature fluctuations, and exposure to harsh elements
- ***Driver integrated circuits (ICs).*** These ICs are pivotal in coordinating and controlling display functions, guaranteeing smooth and efficient operation in synchronisation with the vehicle's electronic systems
- ***Glass moulding.*** Glass moulding processes are employed to shape and form glass components used in displays, requiring precision to achieve the desired optical properties

The evolution of automotive displays extends its influence across various sectors, from enhancing user experiences to creating new revenue streams for manufacturers and suppliers. As vehicles become more connected and sophisticated, the role of automotive displays in shaping the future of mobility cannot be overstated.

vantages extend beyond enhanced brightness and image quality; they are expected to provide substantial power savings, particularly when compared to LCDs. This feature alone could make them a game-changer in the energy-conscious automotive sector.

That said, the microLED industry is still in its growth phase. The ecosystem comprises specialist firms dedicated to microLED technology and larger, established display manufacturers. Collaborations are commonplace, with smaller microLED experts partnering with major display producers. The aim is to refine the technology, ramp up production, and eventually achieve economies of scale.

It is worth noting that incumbent technologies, such as LCDs, have had decades to mature, while OLEDs are approaching a level of maturity after being around for a significant amount of time. In contrast, microLEDs are still in their early stages. Yet, their potential is undeniable, and they are poised to play a pivotal role in redefining automotive displays.

The ultimate success of microLEDs depends on their technological maturity and significant cost reduction. As the technology evolves and prices drop, we may see microLEDs emerge as the new standard in automotive displays.

3D visualisation advances

In the constantly changing world of car screens, light field displays and computer-generated holography are the next big innovations bringing 3D views. These advancements challenge conventional two-dimensional displays, such as LCDs and OLEDs, and pave the way for immersive three-dimensional experiences.

Navigating depth in two dimensions. Light field displays, also known as spatial light field displays, are at the forefront of this revolution. Companies like Leia Inc in the US and Creal in Switzerland are pioneering this technology, which introduces multiple depths to a two-dimensional image. In simpler terms, imagine a screen with 4K pixels; the goal is to make it appear as though there are eight layers within an object. Spatial light field displays achieve this by distributing the pixels evenly among these layers. This allocation of screen resolution to each layer results in accurate depth perception. However, this division of pixels comes at a cost, leading to a reduction in overall resolution.

The motivation behind spatial light field displays is to mitigate the discomfort caused by vergence accommodation conflict, a perceptual challenge in virtual and augmented reality. This conflict arises when our brain perceives an object at one distance while our eyes are focused on a screen at a different distance. Spatial light field displays address this by providing more accurate depth perception, reducing motion sickness, and enhancing the overall experience. Despite being computationally intensive and power-hungry, spatial light field displays are making significant progress, particularly for consumer electronics and applications like vehicle panel side displays. Companies like Leia and FUTURUS are actively collaborating and developing light field displays for commercial use.

Reconstructing reality with precision. Computer-generated holography offers a distinct approach to three-dimensional visualisation. Unlike spatial light field displays,